

Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	P.Swetha nagasri
Student Name	Kallu shresta priya
Roll Number	2520030127
Branch	CSE
Course Outcome	CO-3

Case Study Topic: Internet Backbone – Minimum Cost ISP Peering Plan Using Borůvka's MST Algorithm

Work Done Summary:

In this case study, Borůvka's Minimum Spanning Tree (MST) algorithm was used to compute the minimum-cost ISP backbone network. Each ISP data center was represented as a vertex and peering costs were represented as weighted edges. The algorithm selected the minimum outgoing edge from each component in parallel and merged components until a single spanning tree was formed.

Implementation Details:

- Represented ISP data centers as graph vertices.
- Represented peering links as weighted edges.
- Initialized each vertex as an individual component.
- Applied Borůvka's MST algorithm phase by phase.
- Selected minimum outgoing edge for each component.
- Merged components using Union-Find.
- Repeated until only one component remained.
- Calculated total MST cost.

Result: Screenshots / Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    +

PS C:\Users\kallu shresta priya\Desktop\java> cp 'C:\Users\kallu shresta priya
AppData\Roaming\Code\User\workspaceStorage\da4abde8bf9ecc791a51795e3d48c1dc\
redhat.java\jdt_ws\java_cac89860\bin' 'BoruvkaMST'
MST Edges:
0 - 1 : 4
1 - 2 : 2
3 - 4 : 3
3 - 5 : 4
5 - 6 : 4
5 - 7 : 13
2 - 5 : 6
Total Cost = 36
PS C:\Users\kallu shresta priya\Desktop\java> █
```

Time Complexity

$O(E \log V)$

GitHub Link: <https://github.com/kallushrestapriya-byte/DSA-case-study-3>

Student Signature	Faculty Signature